Why the Universe Disagrees About Its Own Expansion:

Structural Commensurability and Closure-Conditioned Expansion Inference in Darke Theory

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Abstract

Persistent disagreement in inferred cosmic expansion need not imply that one clean scalar expansion history is being imperfectly estimated. It may instead indicate failure of structural commensurability between observational regimes. This paper develops a Darke-theoretic framework in which expansion inference is allowed to depend on direction, redshift, environment, and closure-conditioned legibility. A scalar expansion reduction is defined and shown to be valid only when the residual observational field is structurally null after admissible conditioning. A theorem chain is then established. First, if directional or environment-conditioned residual structure survives redshift-matched and nuisance-controlled reduction, then no purely scalar-commensurate expansion inference is valid on the tested domain. Secondly, if the residual field contains a nonconstant closure-conditioned deformation, then scalar expansion estimators built on different subdomains of the same observational domain need not agree. Thirdly, isotropic scalar recovery is shown to arise only as a limiting case in which directional, environmental, and closure-legibility structure collapse uniformly to isotropic forms. A minimal deformation model is given in which inferred expansion observables acquire explicit dependence on void-conditioned, legibility-conditioned, and directional fields, while the isotropic scalar case appears only when the corresponding deformation coefficients vanish. The observational bridge is supplied by the low-redshift Hubble residual field, for which a nonzero dipole and environment-structured sector asymmetries have already been reported. The paper does not claim final global resolution of all expansion disagreement. Its narrower claim is mathematical and structural: persistent structured disagreement is inconsistent with scalar commensurability and is naturally expected under a universe whose expansion is inferred through structurally non-equivalent cosmological regimes.

1 Introduction

Modern cosmology often compresses late-time expansion inference into one scalar quantity. Operationally, this is convenient. Conceptually, it is powerful. If the universe admits a single clean late-time expansion reading, then a broad range of observational channels should reduce, after admissible conditioning, to one common scalar description. Different datasets may have different error bars, different nuisance structures, and different calibration burdens, but they are still presumed to be reading one structurally commensurate background.

The present paper examines that presumption directly.

The immediate issue is not merely that one famous pair of expansion estimates fails to agree. The deeper issue is that disagreement in inferred expansion may persist in ways that are not well described as random fluctuation around a common scalar truth. At low redshift, the distance-modulus residual field has already been reported to show a fitted dipole component, directional stability across several redshift truncations, and environment-structured sector behaviour linked more strongly to a published local-void field than to a failed CF4/EDD proxy. These features do not by themselves overthrow standard cosmology. They do, however, force a more careful question: under what mathematical conditions is scalar expansion inference valid at all?

That is the problem this paper addresses.

The paper develops its answer inside the Darke Theory framework. Only the load-bearing architectural spine is required:

$$\text{recursive differentiation} \Longrightarrow \text{closure} \Longrightarrow \text{admissibility} \Longrightarrow \text{persistent structure.}$$

This sequence matters because it already distinguishes between formal describability and genuinely supportable structure. A state, field, or observational reduction may be writable without being a true regime invariant. If closure, support, and legibility are not uniformly distributed across observational regimes, then one should not assume that every route to expansion inference is structurally equivalent by default.

The central thesis of the paper is narrow and exact:

> A scalar expansion inference is valid only under structural commensurability. If directional or environment-conditioned residual structure survives admissible conditioning, then one global inferred expansion scalar is not an invariant of the tested observational regime.

This thesis is stronger than a complaint about systematics, but narrower than a wholesale replacement of cosmology. The aim is not to derive a full new cosmological dynamics in the present paper. The aim is to define the exact mathematical point at which scalar

reduction fails, to state the theorem chain associated with that failure, and to show that a Darke-theoretic alternative — closure-conditioned expansion inference — naturally accommodates persistent structured disagreement without requiring that one side of the disagreement be reduced immediately to mere error.

The paper proceeds as follows. Section 2 recalls the minimum foundational setting in Darke Theory required for the argument. Section 3 defines expansion inference and scalar reduction. Section 4 introduces structural commensurability as the criterion governing validity of scalar expansion compression. Section 5 develops the directional and environment-conditioned observational objects used throughout the theorem chain. Section 6 introduces closure-conditioned expansion inference and states a minimal deformation model. Section 7 proves the main theorem chain. Section 8 supplies the low-redshift empirical bridge. Section 9 relates the present construction to wider cosmological non-uniform legibility already developed elsewhere in the Darke Theory corpus. Section 10 states scope, falsifiability, and non-claims. Section 11 closes.

2 Foundational Setting in Darke Theory

The present argument stands downstream of a broader Darke Theory corpus and uses only the minimum load-bearing structure necessary for the current problem. The purpose of this section is not to repeat the entire framework. It is to isolate the exact foundational commitments required for a mathematical treatment of expansion disagreement.

2.1 Recursive differentiation

Darke Theory does not begin from fully furnished objecthood, metric space, or ready-made law acting on already-separated entities. Its starting point is recursive differentiation. Structure is not assumed as a primitive inventory. It is generated by refinement. Distinction is prior to stable identity, and stabilised identity is earned rather than granted at the outset.

The relevant consequence for the present paper is immediate. If structure in general is not primitive, then one should not grant primitive equivalence between all observational regimes. Any claim that multiple cosmological observables must collapse to one common scalar reading already presupposes a stronger uniformity than the foundational setting itself automatically supplies.

2.2 Closure

Differentiation alone does not produce a world. It produces branching or proliferating structural candidates. Closure is the event by which some of those candidates become self-integrating, self-supporting, and persistent. Closure is therefore the threshold between proliferation and stable structure.

This matters for expansion inference because a cosmological observable may be formally available before it is equally legible or equally stabilised across all lines of sight, all redshift sectors, or all environment-conditioned pathways. The present paper treats that possibility seriously.

2.3 Admissibility

Darke Theory distinguishes formal definability from admissible persistence. A regime is not admissible merely because it can be written down. It must be supportable within the burden, closure, and coherence constraints of the governing system. This distinction becomes operational in later Darke-theoretic work through the language of support, burden, lifetime, and admissibility budget.

The present paper uses only the minimal consequence of this distinction: a scalar reduction may be mathematically convenient and still fail to be a true regime invariant if its residual field retains persistent structured content after admissible conditioning.

2.4 Persistent structure

What is real in the strong structural sense is what can be stably carried. Persistent structure is what remains supportable under continued burden, propagation, and refinement. In the present context, this means that a common scalar expansion reduction should not be treated as an invariant merely because it can be fitted. It must also survive tests of structural nullity in its residual field.

2.5 Foundational consequence for expansion inference

These four elements together yield the relevant foundational consequence:

> Darke Theory does not automatically grant structural equivalence between all observational domains. It therefore does not automatically grant that local and deep cosmological inference routes must collapse to one scalar reading in every tested regime.

This is not yet a theorem. It is the correct foundational discipline from which the theorem chain of the paper proceeds.

3 Expansion Inference and Scalar Reduction

The present paper is not about expansion history in the abstract. It is about expansion inference. That distinction is decisive.

3.1 Observational map

Let X denote the class of effective cosmological states, and let $x \in X$ be a fixed effective late-time cosmological state. Let $\hat{n} \in S^2$ denote a line-of-sight direction and let $z \in [0, z_{\max}]$ denote redshift. The observational domain is

$$\Omega \subseteq S^2 \times [0, z_{\max}].$$

Definition 3.1 (Observational map)**.** The observational map associated with state x is a function

$$\mathcal{O}_x : (\hat{n}, z) \mapsto \mathcal{D}_x(\hat{n}, z),$$

where $\mathcal{D}_x(\hat{n}, z)$ is a distance-like observable induced by the cosmological state along line of sight $\hat{n}$ at redshift z.

In a supernova distance setting, the natural realised observable is the distance modulus field, written

$$\mu_x(\hat{n}, z).$$

The reason for beginning this way is that the paper does not identify expansion with any one parameter at the outset. It begins from the field of observables through which expansion is inferred.

3.2 Scalar reduction

A scalar expansion reduction is the claim that the observational map can be compressed to one isotropic background law plus a residual field that is null in the correct structural sense.

Definition 3.2 (Scalar reduction)**.** A scalar reduction on domain Ω is the existence of a scalar parameter H_* and an isotropic background law

$$\mathcal{D}_{\text{iso}}(z; H_*)$$

such that

$$\mathcal{D}_x(\hat{n}, z) = \mathcal{D}_{\text{iso}}(z; H_*) + R_x(\hat{n}, z) \qquad \text{for all } (\hat{n}, z) \in \Omega.$$

Definition 3.3 (Residual field)**.** The residual field associated with a scalar reduction is

$$R_x(\hat{n}, z) := \mathcal{D}_x(\hat{n}, z) - \mathcal{D}_{\text{iso}}(z; H_*).$$

This is the precise mathematical object that carries the entire burden of the paper. Scalar reduction is not under attack simply because it exists. The question is whether the residual field is structurally null after admissible conditioning.

3.3 Low-redshift intercept baseline

For the low-redshift supernova bridge, a natural operational baseline is the intercept-only Hubble form

$$\mu_{\text{model}}(z) = 5 \log_{10}(cz) + A,$$

where A is a fitted intercept parameter. The associated residual field is

$$\Delta\mu(\hat{n}, z) = \mu_{\text{obs}}(\hat{n}, z) - \mu_{\text{model}}(z).$$

The importance of this baseline is not that it is the final cosmographic truth. Its importance is that it provides a disciplined low-redshift reference against which structured directional and environment-conditioned residual content can be tested directly.

3.4 Why scalar reduction must be tested, not assumed

A decomposition into isotropic law plus residual is always mathematically possible once the law family is chosen. That fact alone carries almost no structural content. The true question is whether the residual field can be made structurally null after admissible nuisance conditioning. If not, the existence of the decomposition does not justify scalar compression as a regime invariant. It merely shows that one has performed a fit.

That distinction motivates the central object of the next section: structural commensurability.

4 Structural Commensurability

Scalar reduction is not enough. The same formal decomposition can conceal two very different regimes:

(i) a genuinely scalar-commensurate regime, in which the residual field is nuisance-only after admissible conditioning;

(ii) a non-commensurate regime, in which the residual field retains persistent directional, environment-conditioned, or sector-conditioned structure.

The present paper exists to separate these cases mathematically.

4.1 Structural nullity

We first formalise the idea that a residual field can be null not merely in the sense of having small amplitude, but in the stronger structural sense of lacking persistent organised content after admissible controls.

Definition 4.1 (Structural commensurability)**.** A scalar reduction is structurally commensurate on Ω if, after admissible nuisance conditioning, the associated residual field R_x is structurally null in all of the following senses:

(i) Mean-nullity:

$$\mathbb{E}[R_x(\hat{n}, z) \mid z] = 0.$$

(ii) Directional nullity: the directional harmonic content relevant to the tested scale vanishes under the permitted control ensemble.

(iii) Environment nullity: no admissible environment proxy retains persistent correlation with the residual field after redshift matching and nuisance conditioning.

(iv) Sector nullity: no admissible axis/environment test yields persistent sector asymmetry after redshift matching and nuisance conditioning.

This definition is the hard criterion for when one global inferred expansion scalar can be treated as a regime invariant on the tested domain.

4.2 Why each nullity condition is load-bearing

Mean-nullity alone is weak. A residual field can have conditional mean zero and still carry structured directional or environment-conditioned organisation.

Directional nullity is load-bearing because a persistent large-angle dipole or harmonic mode means the fitted scalar is not invariant under domain restriction. Different directional sectors can then inherit different effective scalar estimates from the same underlying observational field.

Environment nullity is load-bearing because a residual field that remains correlated with void, density-deficit, or other structural proxies is not merely random scatter. It carries

surviving organisation relative to the ambient large-scale structure.

Sector nullity is load-bearing because opposite sectors split relative to a fitted axis precisely when the scalar estimator is becoming subdomain-dependent. At that point, the claim that one clean scalar is being read across the whole tested domain becomes mathematically unstable.

4.3 Closure-conditioned non-commensurability

The paper's alternative is not the denial of scalar reduction in every case. It is the claim that scalar reduction may fail because the residual field contains nontrivial closure-conditioned deformation.

Definition 4.2 (Closure-conditioned non-commensurability)**.** The observational regime is closure-conditioned non-commensurate on Ω if no scalar reduction exists whose residual field is structurally null, and the surviving residual field admits a representation of the form

$$R_x(\hat{n}, z) = \mathcal{F}[E, \Phi](\hat{n}, z) + \epsilon(\hat{n}, z),$$

where:

(i) $\mathcal{F}$ is nonconstant on Ω in direction, environment, or closure-legibility content;

(ii) ϵ is admissible remainder noise or control residue.

This is the paper's formal version of saying that the universe may appear to disagree about its own expansion because expansion is being inferred through structurally non-equivalent cosmological regimes.

5 Directional and Environment-Conditioned Observational Fields

We now introduce the concrete observational objects used in the theorem chain.

5.1 Directional decomposition

A residual field on the sphere naturally admits low-order angular decomposition. The present paper requires only the minimal form:

$$R_x(\hat{n}, z) = a(z) + \mathbf{b}(z) \cdot \hat{n} + Q(\hat{n}, z),$$

where:

- $a(z)$ is an isotropic residual offset,
- $\mathbf{b}(z) \cdot \hat{n}$ is the dipole contribution,
- $Q(\hat{n}, z)$ contains higher-order directional content.

Definition 5.1 (Cumulative dipole amplitude)**.** The cumulative dipole amplitude to upper redshift $z_{\max}$ is

$$B(z_{\max}) := \|\mathbf{b}(z_{\max})\|.$$

Definition 5.2 (Directional non-nullity)**.** A residual field is directionally non-null on Ω if

$$B(z_{\max}) > 0$$

and this persists under the admitted null/control family.

This is the minimal mathematical object needed to formalise a directional late-time Hubble residual field.

5.2 Environment proxies and projected functionals

Let $f(\hat{n}, z)$ be an admissible environment proxy field. Examples include void overlap, density deficit, large-scale stratification proxies, or related structural-conditioning observables.

Let $w(z)$ be an admissible redshift weight. Define the projected line-of-sight environment functional by

$$\mathcal{E}_f(\hat{n}) := \int_0^{z_{\max}} f(\hat{n}, z)\, w(z)\, dz.$$

This converts a line-of-sight environment field into a directly comparable projected quantity at the level of the residual dataset.

5.3 Residual–environment coupling

Definition 5.3 (Residual–environment correlation)**.** For an admissible environment proxy f, define the residual–environment correlation

$$\Gamma_f := \mathrm{Corr}\Big(R_x(\hat{n}_i, z_i), \mathcal{E}_f(\hat{n}_i)\Big).$$

The role of Γ_f is straightforward: if scalar commensurability holds, then no admissible environment proxy should retain persistent residual coupling after admissible conditioning. A nonzero value is therefore a direct indicator of environment non-nullity.

5.4 Sector asymmetries

Let $\hat{d} \in S^2$ be a candidate axis. Define opposite hemisector sets by

$$S_{\pm}(\hat{d}) := \left\{ i : \pm(\hat{n}_i \cdot \hat{d}) > 0 \right\}.$$

Definition 5.4 (Sector environment asymmetry)**.** The sector environment asymmetry relative to axis $\hat{d}$ is

$$\mathcal{A}_f(\hat{d}) := \overline{\mathcal{E}_f}^{(+)} - \overline{\mathcal{E}_f}^{(-)}.$$

Definition 5.5 (Sector residual asymmetry)**.** The sector residual asymmetry relative to axis $\hat{d}$ is

$$\mathcal{R}_{\text{sec}}(\hat{d}) := \overline{R_x}^{(+)} - \overline{R_x}^{(-)}.$$

The sector asymmetry variables matter because they detect domain dependence of the scalar estimator in the most direct observational way. If opposite sectors split systematically relative to a fitted axis, then one common scalar is not being read neutrally across the observational domain.

5.5 Structural question made testable

Once B, Γ_f, $\mathcal{A}_f$, and $\mathcal{R}_{\text{sec}}$ are defined, the validity of scalar reduction ceases to be a matter of informal judgement. It becomes a testable structural question. The paper's theorem chain operates precisely on these objects.

6 Closure-Conditioned Expansion Inference

The present section introduces the Darke-theoretic alternative to automatic scalar commensurability.

6.1 What is being deformed

The paper does not claim that cosmic expansion itself is meaningless. It claims something more disciplined: the observational route from cosmological state to scalar expansion estimator may be structurally deformed by direction, environment, and closure-conditioned legibility.

That means the physical state x may be single while the inferred scalar expansions extracted from different subdomains need not agree.

6.2 General deformation law

Definition 6.1 (Closure-conditioned inference law)**.** A closure-conditioned expansion inference law on Ω is any observational law of the form

$$R_x(\hat{n}, z) = \mathcal{F}[E, \Phi](\hat{n}, z) + \epsilon(\hat{n}, z),$$

where:

(i) $E(\hat{n}, z)$ is an environment field;

(ii) $\Phi(\hat{n}, z)$ is a closure/legibility field;

(iii) $\mathcal{F}$ is a nontrivial deformation operator;

(iv) ϵ is admissible remainder noise.

The meaning is direct. A line of sight need not be observationally neutral with respect to expansion inference if it traverses a structurally non-equivalent environment or a region whose cosmological legibility is not equivalent to that of another line of sight.

6.3 Minimal explicit model

The paper adopts the following minimal residual deformation law:

$$R_x(\hat{n}, z) = \alpha\, V(\hat{n}, z) + \beta\, \Xi(\hat{n}, z) + \gamma\, \hat{n} \cdot \mathbf{d} + \epsilon(\hat{n}, z),$$

where:

- $V(\hat{n}, z)$ is a void- or environment-conditioned field,
- $\Xi(\hat{n}, z)$ is a closure/legibility proxy field,
- $\mathbf{d} \in \mathbb{R}^3$ is a preferred directional mode,
- ϵ is admissible remainder noise.

Definition 6.2 (Minimal closure-conditioned residual model)**.** Any residual law of the form above is called a minimal closure-conditioned residual model.

The scalar-commensurate limit is recovered exactly when

$$\alpha = \beta = \gamma = 0.$$

This is important. The paper does not deny scalar reduction in principle. It states the precise conditions under which scalar reduction is recovered as a limiting case.

6.4 Scalar estimator

The paper also requires a generic scalar estimator notation before the theorem chain proper begins.

Definition 6.3 (Scalar expansion estimator)**.** Let $\Omega' \subseteq \Omega$. A scalar expansion estimator on Ω' is any map

$$\widehat{H}[\Omega']$$

constructed from observational data restricted to Ω', whose output is interpreted as an effective scalar expansion reading for that subdomain.

No uniqueness is assumed. The theorems below require only that the estimator inherit the residual content of the observational field from which it is built.

6.5 Interpretive meaning

The coefficients α, β, γ are not decorative. They state three mathematically distinct ways scalar compression can fail:

(i) V-dependence means residual environment-structured deformation;

(ii) Ξ-dependence means residual closure/legibility-conditioned deformation;

(iii) directional mode dependence means residual preferred-axis deformation.

If any of these survive admissible conditioning, then a scalar estimator built on one subdomain need not agree with the same estimator built on another subdomain of the same cosmological state.

7 Theorem Chain

We now state and prove the paper's main structural results.

7.1 Failure of scalar commensurability under persistent structured residuals

Theorem 7.1 (Failure of scalar commensurability under persistent structured residuals)**.** *Let $R_x(\hat{n}, z)$ be the residual field obtained from an optimal scalar reduction on Ω. Suppose that after admissible nuisance conditioning and redshift-matched controls there exists an admissible environment family f such that*

$$B(z_{\max}) > 0$$

and either

$$\Gamma_f \neq 0$$

or

$$\mathcal{A}_f(\hat{d}) \neq 0$$

for a fitted axis $\hat{d}$. Then the scalar reduction is not structurally commensurate on Ω.

Proof. Structural commensurability requires structural nullity of the residual field. By Definition 4.1, this includes directional nullity and environment/sector nullity after admissible nuisance conditioning. The assumptions of the theorem explicitly violate these requirements: the dipole amplitude remains nonzero, and in addition either the residual field retains nonzero environment correlation or the projected environment field retains nonzero sector asymmetry relative to a fitted axis. Hence the residual field is not structurally null in the sense required for commensurability. Therefore the scalar reduction is not structurally commensurate on Ω. □

Remark 7.2. This theorem is the paper's first exclusion step. It does not assert that every observational disagreement is structural. It states the exact mathematical point at which scalar commensurability fails.

7.2 Closure-conditioned deformation yields subdomain-dependent expansion inference

Theorem 7.3 (Closure-conditioned deformation yields subdomain-dependent expansion inference)**.** *Suppose the residual field admits the representation*

$$R_x(\hat{n}, z) = \mathcal{F}[E, \Phi](\hat{n}, z) + \epsilon(\hat{n}, z),$$

where $\mathcal{F}$ is nonconstant on Ω with respect to direction or admissible environment structure. Let $\widehat{H}[\Omega']$ be any scalar expansion estimator constructed from observational data restricted to subdomain $\Omega' \subseteq \Omega$. Then there exist subdomains $\Omega_1, \Omega_2 \subseteq \Omega$ such that

$$\widehat{H}[\Omega_1] \neq \widehat{H}[\Omega_2]$$

unless $\mathcal{F}$ has identical domain-averaged contribution on Ω_1 and Ω_2.

Proof. Since $\mathcal{F}$ is nonconstant on Ω, there exist at least two nontrivial subdomains Ω_1, Ω_2 for which

$$\langle \mathcal{F} \rangle_{\Omega_1} \neq \langle \mathcal{F} \rangle_{\Omega_2},$$

unless equality is enforced by accidental cancellation or by deliberate construction of subdomains with identical effective averages. The scalar estimator $\widehat{H}[\Omega']$ is built from observables whose residual content includes the deformation term $\mathcal{F}$. Therefore its effective

fitted value acquires dependence on the domain-averaged deformation contribution. Hence distinct subdomains with distinct deformation averages induce distinct inferred scalar values:

$$\widehat{H}[\Omega_1] \neq \widehat{H}[\Omega_2].$$

Equality can occur only if the domain-averaged deformation contribution is identical on the two subdomains. □

Remark 7.4. This theorem earns the title of the paper. One underlying cosmological state may yield non-equivalent inferred scalar expansion readings whenever the observational inference law contains nonconstant structural deformation.

7.3 Isotropic recovery as a limiting case

Theorem 7.5 (Isotropic scalar recovery as a limiting case)**.** *Assume that on* Ω*,*

$$E(\hat{n}, z) \to E_0(z), \qquad \Phi(\hat{n}, z) \to \Phi_0(z)$$

uniformly, and additionally

$$\nabla_{\hat{n}} \Phi(\hat{n}, z) \to 0.$$

Let $\mathcal{F}[E, \Phi]$ *be any smooth closure-conditioned deformation operator. Then*

$$\mathcal{F}[E, \Phi](\hat{n}, z) \to F_0(z)$$

for some isotropic function F_0*, and scalar commensurability is recovered up to isotropic renormalisation.*

Proof. Uniform convergence of the environment field and closure/legibility field to isotropic z-dependent limits eliminates all persistent directional and environment-conditioned variation entering the deformation operator. Since $\mathcal{F}$ is smooth in its arguments, its dependence on angularly varying structure collapses to dependence only on the limiting isotropic fields. Therefore there exists a function $F_0(z)$ such that

$$\mathcal{F}[E, \Phi](\hat{n}, z) \to F_0(z).$$

The residual field therefore differs from an isotropic scalar reduction only by an isotropic renormalisation contribution. Directional nullity, environment nullity, and sector nullity are recovered in the limit, so structural commensurability is restored. □

Remark 7.6. This theorem is essential for discipline. The paper is not an anti-isotropy manifesto. It states that scalar commensurability is a limiting structural case, not an automatic ontological default.

7.4 Minimal-model corollary

Corollary 7.7 (Minimal-model commensurate limit)**.** *For the minimal closure-conditioned residual model*

$$R_x(\hat{n}, z) = \alpha\, V(\hat{n}, z) + \beta\, \Xi(\hat{n}, z) + \gamma\, \hat{n} \cdot \mathbf{d} + \epsilon(\hat{n}, z),$$

scalar commensurability is recovered if and only if the deformation coefficients satisfy

$$\alpha = \beta = \gamma = 0$$

up to admissibly null isotropic renormalisation.

Proof. If $\alpha = \beta = \gamma = 0$, the residual field reduces to admissible remainder noise, possibly plus isotropic renormalisation. Structural nullity is then compatible with the definitions of scalar commensurability and closure-conditioned non-commensurability. Conversely, if any of α, β, γ is nonzero and the corresponding field survives admissible conditioning, then directional, environment, or closure-conditioned structure remains, which violates structural nullity and therefore structural commensurability. □

8 Empirical Bridge from the Low-Redshift Hubble Residual Field

The theorem chain above is general. It does not depend on any one observational dataset. The purpose of the present section is to show that the formal objects developed in the paper already have a concrete empirical bridge.

8.1 Reported low-redshift directional structure

In the low-redshift Pantheon+SH0ES residual analysis over

$$0.01 \leq z_{\mathrm{CMB}} \leq 0.15,$$

the fitted dipole model yields a residual dipole amplitude of

$$B = 0.080799 \text{ mag},$$

with axis

$$(\mathrm{RA}, \mathrm{Dec}) = (23.331^\circ, 37.617^\circ).$$

The same analysis reports that:

(i) the dipole survives shuffle-based null tests;

(ii) the cumulative low-redshift directional component remains stable under several redshift truncations;

(iii) custom environment and stratification proxies show structured projection-side and sector-side behaviour;

(iv) a CF4/EDD-based proxy fails to independently replicate the result;

(v) a published local-void catalogue does independently replicate the sector-based environmental distinction.

These points matter because the theorem chain of the present paper does not require perfect global replication in every proxy. It requires only that admissible directional and environment-conditioned non-nullity survive under disciplined controls. That is precisely the kind of empirical structure these reported results instantiate.

8.2 Empirical proposition

Proposition 8.1 (Low-redshift empirical bridge instance)**.** *On the reference low-redshift supernova sample* $0.01 \leq z_{\mathrm{CMB}} \leq 0.15$*, the reported fitted residual dipole amplitude*

$$B = 0.080799 \text{ mag}$$

with axis

$$(\mathrm{RA}, \mathrm{Dec}) = (23.331^\circ, 37.617^\circ),$$

together with reported environment-structured sector behaviour and published-void replication, constitutes an empirical bridge instance of directional and environment-conditioned residual non-nullity in the sense of Sections 4 and 5.

Proof. A nonzero dipole amplitude satisfying the stated fitted residual model is, by definition, directional non-nullity. The persistence of structured sector behaviour with respect to admissible environment proxies establishes environment non-nullity in the operational sense formalised by Γ_f, $\mathcal{A}_f$, and $\mathcal{R}_{\mathrm{sec}}$. Published-void replication further shows that the sector/environment structure is not exhausted by one custom proxy alone. Hence the reported observational field furnishes an empirical example of the non-null residual structure formalised in the present paper. □

8.3 What this bridge does and does not establish

The proposition above does not prove the full Darke-theoretic cosmological interpretation by itself. It does not show that every expansion disagreement is closure-conditioned, nor

does it show that all standard nuisance interpretations are exhausted.

What it does establish is narrower and sufficient for the present paper: the mathematical objects used in the theorem chain are not empty abstractions. They already occur in a live low-redshift residual field.

More precisely, the low-redshift bridge is sufficient to instantiate the variables B, Γ_f, and $\mathcal{A}_f$ at the level of an empirical residual analysis, but it is not sufficient by itself to determine a unique deformation operator $\mathcal{F}$. That distinction is important. The theorem chain requires structured non-nullity; it does not require a unique completed micro-model of the deformation field.

That is enough to make the paper more than a conceptual note. It becomes a theorem-bearing whitepaper with a direct empirical wedge.

9 Relation to Broader Cosmological Legibility Non-Uniformity

Remark 9.1. Section 9 is not evidentiary support for the theorem chain proved in Section 7. Its role is narrower. It shows that closure-conditioned expansion inference belongs to a regime class already admissible elsewhere in the broader Darke Theory cosmology corpus. The section therefore supplies admissibility and framework continuity, not an independent empirical proof of the present theorem chain.

The present paper is not an isolated reinterpretation. It sits naturally inside a wider group of Darke-theoretic cosmology-facing results in which cosmological legibility is not assumed to be uniformly given.

9.1 Preconditioned transparency

In the reionisation whitepaper, radiative transparency is treated not merely as the switching-on of sources in an otherwise uniform medium, but as the unveiling of a universe whose transparency field may already have become unevenly structured. This distinction is important because it already introduces a cosmological setting in which legibility is not globally neutral by default.

The present paper uses an analogous but later-time logic. The claim is not that reionisation and expansion disagreement are the same phenomenon. The claim is that non-uniform cosmological legibility is already a disciplined Darke-theoretic category. If transparency can be preconditioned, then it is mathematically reasonable that expansion inference may also be conditioned by structurally non-equivalent observational sectors.

9.2 Bounded transition-window structure

The Little Red Dots whitepaper places unusually compact high-redshift populations into a bounded transition epoch and interprets them as closure-window structure rather than as smooth standard-hierarchy products everywhere at once. Again, the precise mechanism differs from the present paper, but the structural consequence is similar: cosmological structure may mature unevenly, in bounded windows, rather than everywhere equally and continuously.

This supports the present paper's refusal to assume that local and deep cosmological inference pathways are automatically structurally commensurate.

9.3 Thresholded early-universe emergence

The refinement-instability inflation-replacement work places early cosmological expansion in a thresholded structural setting rather than in a picture of primitive furnished geometry with a later correction applied. This matters because it further supports the idea that large-scale cosmological structure should not be treated as fully uniform and already-legible by default.

9.4 Formal bridge to the present paper

The present section should not remain purely thematic. Its formal point is the following: if non-uniform cosmological legibility is already admissible elsewhere in the Darke Theory cosmology corpus, then closure-conditioned expansion inference is mathematically admissible here as a regime hypothesis. In other words, the current paper does not ask the reader to accept an alien category. It asks the reader to permit, at late times, a structural non-equivalence already admitted elsewhere in the same broader framework.

9.5 Summary of relation

Taken together, these wider cosmology-facing papers do not prove the present claim directly. What they do supply is a broader consistency environment:

(i) cosmological legibility need not be globally uniform;

(ii) transition windows can matter;

(iii) structural readiness and observational access can be non-equivalent across sectors.

This is not introduced here as substitute evidence for the theorem chain. It is introduced to show that the regime class invoked by the present paper is not foreign to the wider

Darke-theoretic cosmology architecture. The current whitepaper is therefore not an ad hoc patch inserted into one anomaly. It is a late-time observational inference application of a broader non-uniform legibility logic already present elsewhere in the corpus.

10 Scope, Falsifiability, and Non-Claims

A structural paper of this kind must state clearly what it is, what it is not, and what would count against it.

10.1 What the paper claims

The paper claims the following:

(i) a scalar expansion inference is valid only under structural commensurability;

(ii) persistent directional and environment-conditioned residual structure mathematically excludes structural commensurability on the tested domain;

(iii) closure-conditioned deformation gives a natural mechanism by which one cosmological state can yield multiple non-equivalent scalar expansion estimates on different subdomains;

(iv) isotropic scalar recovery is a limiting case rather than an automatic starting assumption.

10.2 What the paper does not claim

The paper does not claim:

(i) that the full global expansion disagreement is already completely resolved;

(ii) that one low-redshift anisotropy result alone overturns standard cosmology;

(iii) that every residual or calibration issue is necessarily structural;

(iv) that isotropic scalar cosmology is impossible in principle;

(v) that the minimal closure-conditioned model given here is the unique admissible model.

The paper is narrower. It identifies the exact mathematical condition under which scalar compression fails as a regime invariant.

10.3 What would count against the framework

The present framework would be weakened or falsified on the tested domain if one or more of the following were robustly established:

(i) directional residual structure collapses to null under stronger admissible controls;

(ii) environment-conditioned couplings vanish under properly redshift-matched and nuisance-controlled tests;

(iii) independent replications of the low-redshift directional/environment structure fail in a way that leaves only unstable or proxy-specific artefacts;

(iv) no environment or closure-legibility proxy improves explanatory organisation over ordinary nuisance treatment;

(v) subdomain-dependent scalar estimates collapse to a common value once admissible structural conditioning is correctly included, thereby removing the need for closure-conditioned non-commensurate deformation on the tested domain.

10.4 Why these limits strengthen rather than weaken the paper

The present scope discipline is not a concession. It is required by the broader Darke-theoretic architecture itself. A theory concerned with admissibility and bounded structural support must state where its reasoning applies and where it does not. The current whitepaper applies only where directional, environment-conditioned, or legibility-conditioned residual structure can be defined with enough precision to render structural commensurability a real mathematical question.

11 Conclusion

This paper has not attempted to replace cosmology with a complete new total system. It has attempted something narrower and more exact.

A scalar expansion inference is valid only under structural commensurability. If the residual field remaining after scalar reduction is structurally null after admissible conditioning, then one global inferred expansion parameter may be treated as a regime invariant on the tested domain. If, however, directional and environment-conditioned residual structure survives admissible conditioning, then scalar commensurability fails.

That result was formalised in three steps. First, persistent structured residuals exclude structural commensurability. Secondly, nonconstant closure-conditioned deformation implies subdomain-dependent scalar expansion inference even for one underlying cosmological

state. Thirdly, isotropic scalar recovery arises only when directional, environmental, and closure-legibility structure collapse uniformly to isotropic limits.

The paper then supplied a concrete empirical bridge from the low-redshift Hubble residual field, where a nonzero dipole and environment-structured sector behaviour have already been reported. That bridge does not by itself settle the full cosmological interpretation. It does show that the mathematical objects used here are already active in real observational analysis.

The final claim is exact and limited:

> The universe may appear to disagree about its own expansion not because one clean scalar truth is being estimated imperfectly, but because expansion is being inferred through structurally non-equivalent cosmological regimes.

That is the structural result established here.